SET 63 • HSS-ID.B.6.a

BACKWARD 1

Describe patterns in scatter plots • Grade 8 • Statistics & Probability • 8.SP.A.1

Points rise from left to right, close to a line.

Describe the association.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 63 • BACKWARD 1 • Q1/5

What is an outlier in a scatter plot?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 63 • BACKWARD 1 • Q2/5

Points fall from left to right in a curved pattern.

Which describes the association?

A. Positive and nonlinear

B. Positive and linear

C. Negative and nonlinear

D. No association

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 63 • BACKWARD 1 • Q3/5

True or false?

A scatter plot shows the relationship between two quantitative variables.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 63 • BACKWARD 1 • Q4/5

True or false?

If the points show no pattern, there is a strong association.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 63 • BACKWARD 1 • Q5/5

SET 63 • HSS-ID.B.6.a

BACKWARD 2

Write linear and exponential functions through given points • High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2

Write the exponential function through (0, 100) and (1, 130).

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B2-Q1

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 63 • BACKWARD 2 • Q1/5

Write the linear function through (0, 2) and (4, 26).

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B2-Q2

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 63 • BACKWARD 2 • Q2/5

Which exponential function $y = ab^x$ $(b > 0)$ passes through $(0, 50)$ and $(2, 12.5)$?

A. $y = 50(0.25)^x$

B. $y = 12.5(2)^x$

C. $y = 50(0.5)^x$

D. $y = 50 - 18.75x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B2-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 63 • BACKWARD 2 • Q3/5

True or false?

$y = 3(2)^x$ passes through (2, 12).

A. True

B. False

True or false?

$y = 5(1.2)^x$ passes through (0, 6).

A. True

B. False

SET 63 • HSS-ID.B.6.a

FORWARD 1

Use logarithms with an exponential model fitted to data • High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

$P(t) = 100(1.3)^t$.

When does P reach 1,000? Round to the nearest tenth.

$t \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S63-F1-Q1**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 63 • FORWARD 1 • Q1/5

Solve $2(1.5)^x = 30$. Round to the nearest hundredth.

$x \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F1-Q2

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 63 • FORWARD 1 • Q2/5

Which is the exact solution of $100(1.3)^t = 500$? (log means base 10.)

A. t = log 1.3 ÷ log 5

B. t = 500 ÷ 130

C. t = log 5 ÷ log 1.3

D. t = 5 ÷ 1.3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S63-F1-Q3**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 63 • FORWARD 1 • Q3/5

True or false?

If $1.3^t = 1.69$, then $t = 2$.

A. True

B. False

True or false?

If $10^t = 50$, then $t = 5$.

A. True

B. False

SET 63 • HSS-ID.B.6.a

FORWARD 2

Fit a sinusoidal model to periodic data • High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5

Monthly average temperatures peak at 30 °C and bottom out at 10 °C.

What amplitude and midline fit a sinusoidal model?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 63 • FORWARD 2 • Q1/5

What period fits monthly temperatures that repeat every year?

Period: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F2-Q2

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 63 • FORWARD 2 • Q2/5

Daylight hours vary from 9 to 15 over a year.

Which model fits?

A. A linear model

B. A sinusoidal model with amplitude 3 and period 12 months

C. A quadratic model with vertex at month 12

D. An exponential model

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S63-F2-Q3**

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 63 • FORWARD 2 • Q3/5

True or false?

A repeating pattern in data suggests a sinusoidal model.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 63 • FORWARD 2 • Q4/5

True or false?

Data that vary between 9 and 15 have amplitude 6.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 63 • FORWARD 2 • Q5/5

SET 83 • HSS-ID.B.6.a • INTERPRETING CATEGORICAL & QUANTITATIVE DATA

Construct scatter plots from paired data and describe how the variables are related

MAIN	Algebra I main question	**HSS-ID.B.6.a**
BACKWARD 1	Make and describe scatter plots	**8.SP.A.1**
BACKWARD 2	Plot points and find distances on the coordinate plane	**6.NS.C.8**
FORWARD 1	Choose an exponential model from a scatter plot and solve it with…	**HSF-LE.A.4**
FORWARD 2	Evaluate claims based on scatter plots	**HSS-IC.B.6**

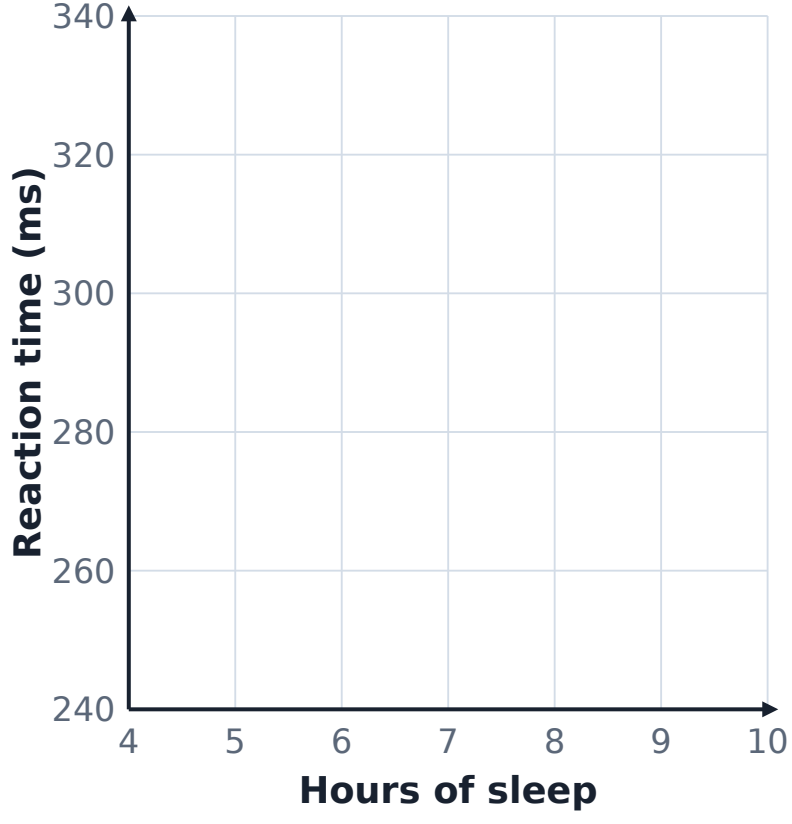
SET 83 • HSS-ID.B.6.a

MAIN

Algebra I main question • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a

Make a scatter plot of hours of sleep (x) and reaction time in milliseconds (y):

(5, 320), (6, 300), (6.5, 290), (7, 280), (8, 260), (9, 255)



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-M-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 83 • MAIN • Q1/5

Describe the association between hours of sleep and reaction time in the data (5, 320), (6, 300), (6.5, 290), (7, 280), (8, 260), (9, 255): its direction, form, and strength.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-M-Q2

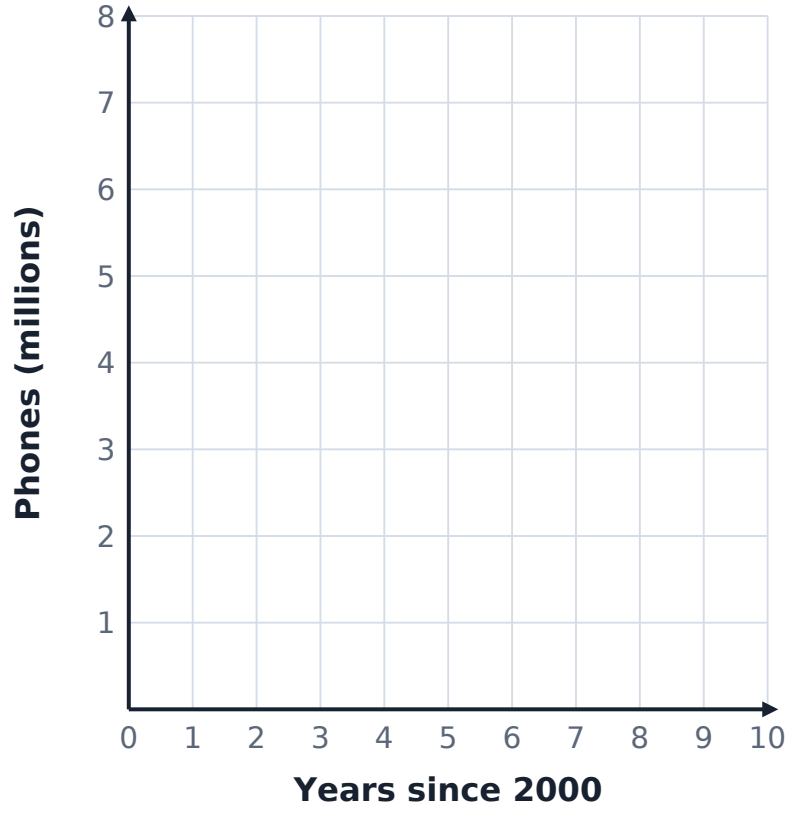
High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 83 • MAIN • Q2/5

Make a scatter plot of years since 2000 (x) and phones in millions (y):

(0, 1), (2, 1.5), (4, 2.3), (6, 3.4), (8, 5.1), (10, 7.6)

Which type of model does the plot suggest?

Model: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-M-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 83 • MAIN • Q3/5

When you make a scatter plot to see whether one variable helps explain another, which variable usually goes on the horizontal axis?

A. Whichever variable has larger numbers

B. Whichever variable has fewer values

C. The explanatory variable (the one that may explain the other)

D. The response variable

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-M-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 83 • MAIN • Q4/5

True or false?

Both axes of a scatter plot must start at 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-M-Q5

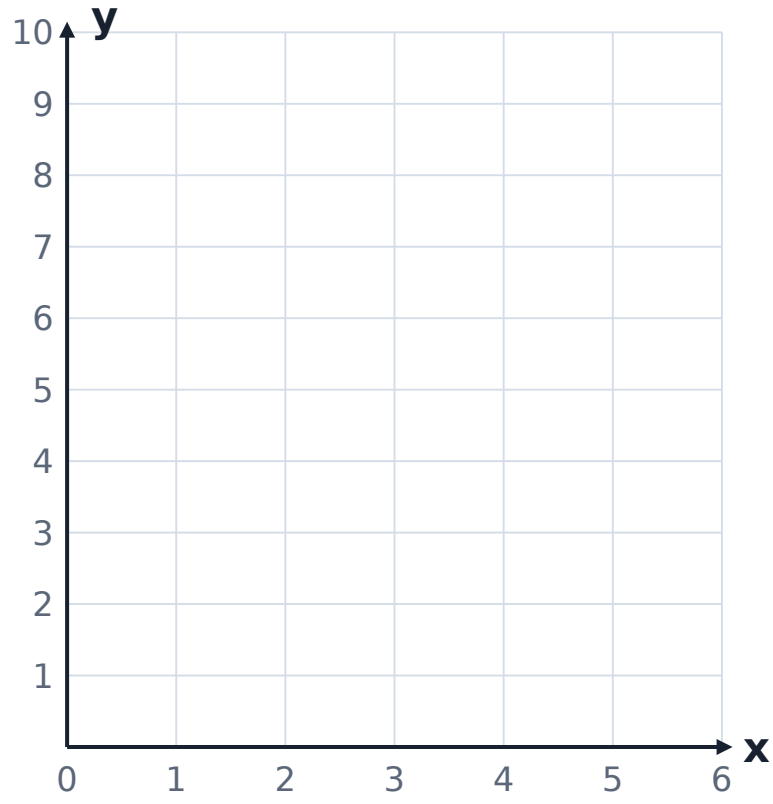
High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 83 • MAIN • Q5/5

SET 83 • HSS-ID.B.6.a

BACKWARD 1

Make and describe scatter plots • Grade 8 • Statistics & Probability • 8.SP.A.1

Plot (1, 2), (2, 3), (3, 5), (4, 6), and (5, 8) as a scatter plot.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-B1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 83 • BACKWARD 1 • Q1/5

Points rise from left to right but curve upward. Describe the association.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-B1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 83 • BACKWARD 1 • Q2/5

Which describes a scatter plot whose points show no trend?

A. Linear association

B. Strong positive association

C. Strong negative association

D. No association

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-B1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 83 • BACKWARD 1 • Q3/5

True or false?

Each point in a scatter plot represents one pair of values.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-B1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 83 • BACKWARD 1 • Q4/5

True or false?

A cluster in a scatter plot is the same thing as an outlier.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-B1-Q5

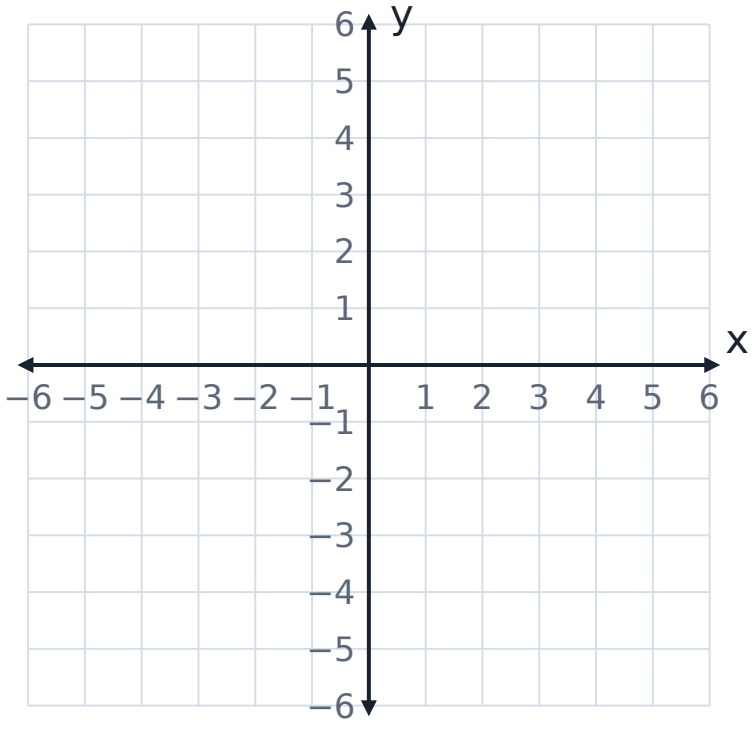
Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 83 • BACKWARD 1 • Q5/5

SET 83 • HSS-ID.B.6.a

BACKWARD 2

Plot points and find distances on the coordinate plane • Grade 6 • The Number System • 6.NS.C.8

Plot the points (2, 5), (−3, 1), and (0, −4).



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-B2-Q1

Grade 6 • The Number System • 6.NS.C.8 • Set 83 • BACKWARD 2 • Q1/5

Find the distance between $(-3, 2)$ and $(5, 2)$.

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-B2-Q2

Grade 6 • The Number System • 6.NS.C.8 • Set 83 • BACKWARD 2 • Q2/5

Which point lies on the y-axis?

A. $(1, -1)$

B. $(0, -4)$

C. $(-4, 0)$

D. $(4, 4)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-B2-Q3

Grade 6 • The Number System • 6.NS.C.8 • Set 83 • BACKWARD 2 • Q3/5

True or false?

The point $(-2, 6)$ is in Quadrant II.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-B2-Q4

Grade 6 • The Number System • 6.NS.C.8 • Set 83 • BACKWARD 2 • Q4/5

True or false?

The distance between $(1, -3)$ and $(1, 4)$ is 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-B2-Q5

Grade 6 • The Number System • 6.NS.C.8 • Set 83 • BACKWARD 2 • Q5/5

SET 83 • HSS-ID.B.6.a

FORWARD 1

Choose an exponential model from a scatter plot and solve it with logarithms • High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

A model for the phone data is $y = 1(1.22)^x$, where x is years since 2000 and y is millions of phones. When does it predict 20 million? Round to the nearest tenth. (log means base 10.)

$x \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-F1-Q1

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 83 • FORWARD 1 • Q1/5

Using $y = 1(1.22)^x$, how many years does it take the number of phones to double? Round to the nearest tenth. (log means base 10.)

Years: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-F1-Q2

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 83 • FORWARD 1 • Q2/5

Solve $4(1.25)^x = 20$. (log means base 10.)

A. $x = \log 5 \div \log 1.25 \approx 7.21$

B. $x = 5 \div 1.25$

C. $x = 4$

D. $x = \log 20 \div \log 1.25 \approx 13.43$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S83-F1-Q3**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 83 • FORWARD 1 • Q3/5

True or false?

If $y = a \cdot b^x$ with $a > 0$ and $b > 1$, then y doubles every $\log 2 \div \log b$ units of x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 83 • FORWARD 1 • Q4/5

True or false?

The model $y = 1(1.22)^x$ predicts 0 phones in 2000.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 83 • FORWARD 1 • Q5/5

SET 83 • HSS-ID.B.6.a

FORWARD 2

Evaluate claims based on scatter plots • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6

A website plots shoe size against reading score for students in Grades 1 to 8, finds a strong positive association, and claims that bigger feet improve reading. What is wrong?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 83 • FORWARD 2 • Q1/5

A scatter plot of 6 students from one class shows a strong trend. Why should you be careful applying it to all students?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 83 • FORWARD 2 • Q2/5

Which detail would make a claim based on a scatter plot more trustworthy?

A. Both axes start at 0.

B. The plot uses bright colors.

C. There are only 5 points.

D. The data come from a large random sample.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S83-F2-Q3**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 83 • FORWARD 2 • Q3/5

True or false?

Extrapolating a trend far beyond the plotted data can give misleading predictions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-F2-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 83 • FORWARD 2 • Q4/5

True or false?

If a scatter plot shows a strong trend, one variable causes the other.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S83-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 83 • FORWARD 2 • Q5/5

SET 64 • HSS-ID.B.6.b • INTERPRETING CATEGORICAL & QUANTITATIVE DATA

Assess the fit of a function by plotting and analyzing residuals

MAIN	Algebra I main question	**HSS-ID.B.6.b**
BACKWARD 1	Use a fitted function to make predictions	**HSS-ID.B.6.a**
BACKWARD 2	Subtract rational numbers	**7.NS.A.1.c**
FORWARD 1	Use logarithms to solve exponential models fitted to data	**HSF-LE.A.4**
FORWARD 2	Evaluate reports that use fitted models	**HSS-IC.B.6**

SET 64 • HSS-ID.B.6.b

MAIN

Algebra I main question • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.b

The model $\hat{y} = 2.2x + 0.8$ is fitted to the data (1, 3), (2, 5), (3, 8), (4, 9), (5, 12).

Find the residual for each point.

Residuals: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-M-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.b • Set 64 • MAIN • Q1/5

For that model, what does the residual -0.6 at $x = 4$ mean?

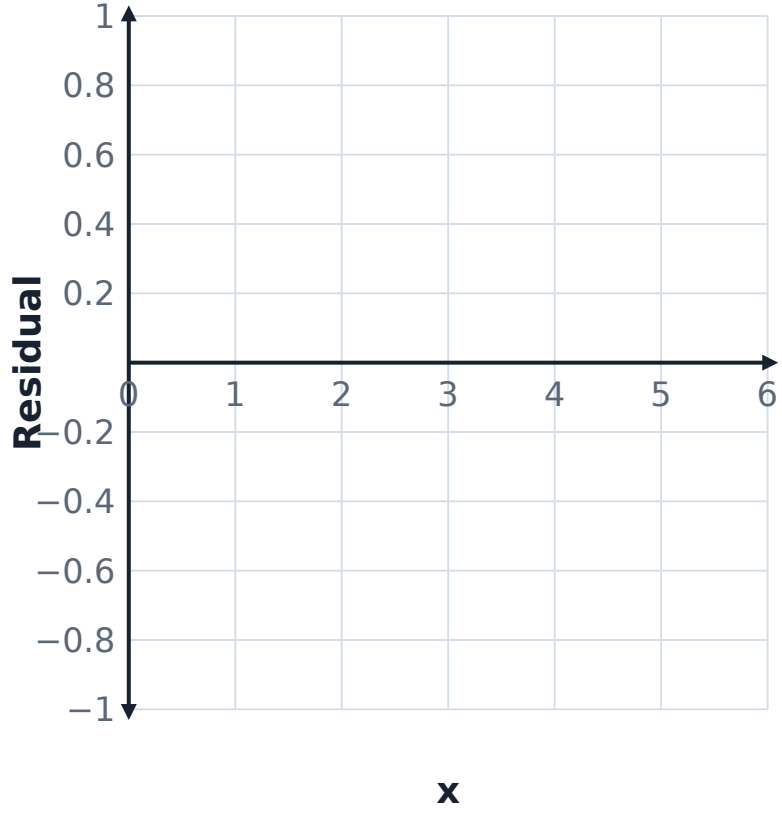
Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-M-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.b • Set 64 • MAIN • Q2/5

Make a residual plot for the residuals 0, −0.2, 0.6, −0.6, 0.2 at x = 1, 2, 3, 4, 5.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-M-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.b • Set 64 • MAIN • Q3/5

A residual plot shows a clear U-shaped pattern.

What does that suggest?

A. The correlation is exactly 1.

B. A linear model is not a good fit; a curved model may fit better.

C. The linear model fits well.

D. There are no residuals.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-M-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.b • Set 64 • MAIN • Q4/5

True or false?

A good linear fit has residuals scattered randomly around 0, with no pattern.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-M-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.b • Set 64 • MAIN • Q5/5

SET 64 • HSS-ID.B.6.b

BACKWARD 1

Use a fitted function to make predictions • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a

The model is $\hat{y} = 2.2x + 0.8$.

Predict y when $x = 6$.

$\hat{y} =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B1-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 64 • BACKWARD 1 • Q1/5

The model is $y = 3x^2 - 2$.

Predict y when x = 4.

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B1-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 64 • BACKWARD 1 • Q2/5

The model is $y = 50(0.9)^x$.

Predict y when x = 2.

A. 50

B. 40.5

C. 45

D. 90

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B1-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 64 • BACKWARD 1 • Q3/5

True or false?

A predicted value comes from the model, not from the data table.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B1-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 64 • BACKWARD 1 • Q4/5

True or false?

If a model predicts 14 and the actual value is 14, the residual is 14.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B1-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 64 • BACKWARD 1 • Q5/5

SET 64 • HSS-ID.B.6.b

BACKWARD 2

Subtract rational numbers • Grade 7 • The Number System • 7.NS.A.1.c

Find $8 - 7.4$.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B2-Q1

Grade 7 • The Number System • 7.NS.A.1.c • Set 64 • BACKWARD 2 • Q1/5

Find $9 - 9.6$.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B2-Q2

Grade 7 • The Number System • 7.NS.A.1.c • Set 64 • BACKWARD 2 • Q2/5

Find $5 - 5.2$.

A. -10.2

B. 10.2

C. 0.2

D. -0.2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B2-Q3

Grade 7 • The Number System • 7.NS.A.1.c • Set 64 • BACKWARD 2 • Q3/5

True or false?

$3 - (-2) = 5$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B2-Q4

Grade 7 • The Number System • 7.NS.A.1.c • Set 64 • BACKWARD 2 • Q4/5

True or false?

$-4 - 6 = -2$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B2-Q5

Grade 7 • The Number System • 7.NS.A.1.c • Set 64 • BACKWARD 2 • Q5/5

SET 64 • HSS-ID.B.6.b

FORWARD 1

Use logarithms to solve exponential models fitted to data • High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

A model fitted to bacteria counts is $N = 150(1.4)^t$, with t in hours. When does N reach 1,000? Give the exact expression and the value to the nearest tenth of an hour. (log means base 10.)

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F1-Q1

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 64 • FORWARD 1 • Q1/5

A line fitted to the logarithms of data is log y = 0.3x + 1 (log means base 10).

Write y as an exponential function of x.

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F1-Q2

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 64 • FORWARD 1 • Q2/5

A fitted model is $y = 20e^{0.08x}$. For which x is y = 60? (ln means base e.)

A. $x = 3 \div 0.08 = 37.5$

B. $x = \ln 60 \div 0.08 \approx 51.2$

C. $x = 0.08 \ln 3 \approx 0.09$

D. $x = \ln 3 \div 0.08 \approx 13.7$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S64-F1-Q3**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 64 • FORWARD 1 • Q3/5

True or false?

For the model $y = 5(1.5)^x$, the solution of $5(1.5)^x = 50$ is $x = \log 10 \div \log 1.5 \approx 5.68$. (log means base 10.)

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 64 • FORWARD 1 • Q4/5

True or false?

The doubling time of $N = 150(1.4)^t$ is $\log 2 \div \log 1.4 \approx 0.49$ hour. (log means base 10.)

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 64 • FORWARD 1 • Q5/5

SET 64 • HSS-ID.B.6.b

FORWARD 2

Evaluate reports that use fitted models • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6

A report fits a line to the heights of children aged 5 to 15 and uses it to predict the height of a 40-year-old.

What is the problem?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions HSS-IC.B.6 • Set 64 • FORWARD 2 • Q1/5

A report finds a strong correlation between ice cream sales and drownings and concludes that ice cream causes drowning.

Do these observational data establish that conclusion? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions HSS-IC.B.6 • Set 64 • FORWARD 2 • Q2/5

A residual plot shows a strong curve, but a report says the linear model is excellent.

What should you conclude?

A. The report is correct.

B. The data must be wrong.

C. The residuals are all 0.

D. The linear model may not be appropriate.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S64-F2-Q3**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions HSS-IC.B.6 • Set 64 • FORWARD 2 • Q3/5

True or false?

A report should state how its data were collected.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F2-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 64 • FORWARD 2 • Q4/5

True or false?

A model with a high correlation coefficient is always the right type of model.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions HSS-IC.B.6 • Set 64 • FORWARD 2 • Q5/5

SET 65 • HSS-ID.B.6.c • INTERPRETING CATEGORICAL & QUANTITATIVE DATA

Fit a linear function to a scatter plot that suggests a linear association

MAIN	Algebra I main question	**HSS-ID.B.6.c**
BACKWARD 1	Judge a line of fit informally	**8.SP.A.2**
BACKWARD 2	Write a line through two points	**HSF-LE.A.2**
FORWARD 1	Evaluate conclusions drawn from a line of fit	**HSS-IC.B.6**
FORWARD 2	Use a randomized experiment and simulation to test whether mo…	**HSS-IC.B.5**

SET 65 • HSS-ID.B.6.c

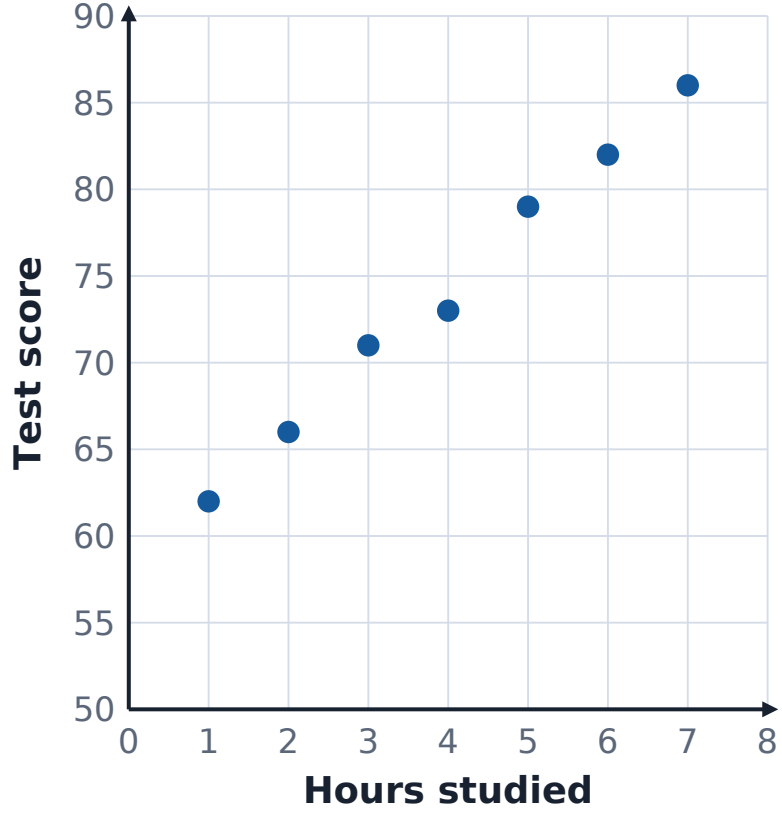
MAIN

Algebra I main question • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c

The scatter plot shows hours studied and test scores.

Draw a line of fit and write its equation.

Equation: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-M-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c • Set 65 • MAIN • Q1/5

The least-squares line for the data is $\hat{y} = 4x + 58.14$.

Predict the score for 4.5 hours, to the nearest whole number.

Score: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-M-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c • Set 65 • MAIN • Q2/5

Write the equation of the line through the data points (1, 62) and (7, 86), to use as a line of fit.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-M-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c • Set 65 • MAIN • Q3/5

Which line fits the points (0, 1), (1, 3), (2, 5), (3, 7) exactly?

A. $y = 3x + 1$

B. $y = 2x + 1$

C. $y = 2x$

D. $y = x + 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-M-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c • Set 65 • MAIN • Q4/5

True or false?

A line of fit should pass through every data point.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-M-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c • Set 65 • MAIN • Q5/5

SET 65 • HSS-ID.B.6.c

BACKWARD 1

Judge a line of fit informally • Grade 8 • Statistics & Probability • 8.SP.A.2

Is a straight line a good model for points that curve strongly upward? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 65 • BACKWARD 1 • Q1/5

How can you judge whether a line of fit is good?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 65 • BACKWARD 1 • Q2/5

Which is the best sign of a good line of fit?

A. All of the points are below it.

B. All of the points are above it.

C. The points are close to the line and follow its trend.

D. Exactly two points are on it.

True or false?

Lines of fit are used for data with a linear association.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 65 • BACKWARD 1 • Q4/5

True or false?

A line of fit must pass through the origin.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 65 • BACKWARD 1 • Q5/5

SET 65 • HSS-ID.B.6.c

BACKWARD 2

Write a line through two points • High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2

Write the equation of the line through (2, 66) and (6, 82).

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B2-Q1

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 65 • BACKWARD 2 • Q1/5

Write the equation of the line through (0, 7) and (3, 13).

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B2-Q2

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 65 • BACKWARD 2 • Q2/5

Which line passes through (0, 10) and (5, 0)?

A. $y = 2x + 10$

B. $y = 10x - 2$

C. $y = -2x + 10$

D. $y = -0.5x + 10$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B2-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 65 • BACKWARD 2 • Q3/5

True or false?

The line through (0, 3) with slope 4 is $y = 4x + 3$.

A. True

B. False

True or false?

The line through (2, 2) and (4, 8) has slope 2.

A. True

B. False

SET 65 • HSS-ID.B.6.c

FORWARD 1

Evaluate conclusions drawn from a line of fit • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6

In a study of 7 volunteer students, each extra hour of study went with about 4 more points.

Can a school claim this holds for all of its students? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-F1-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions

HSS-IC.B.6 • Set 65 • FORWARD 1 • Q1/5

A study of 7 students who studied from 1 to 7 hours found the line $\hat{y} = 4x + 58$, where x is hours of study and $\hat{y}$ is the test score out of 100.

A report uses this line to predict a score of 138 for 20 hours of study. Why is that unreasonable?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-F1-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions HSS-IC.B.6 • Set 65 • FORWARD 1 • Q2/5

Which detail makes a regression conclusion more trustworthy?

A. The prediction is far outside the data.

B. The line was drawn by hand.

C. The data came from a large random sample.

D. The sample was 5 volunteers.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S65-F1-Q3**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 65 • FORWARD 1 • Q3/5

True or false?

Observational data alone cannot prove that one variable causes another.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-F1-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 65 • FORWARD 1 • Q4/5

True or false?

A line of fit from 7 volunteer students proves that extra study raises the score of every student.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-F1-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 65 • FORWARD 1 • Q5/5

SET 65 • HSS-ID.B.6.c

FORWARD 2

Use a randomized experiment and simulation to test whether more study raises scores • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5

To test whether extra study raises scores, 40 volunteers were randomly assigned: 20 studied 3 extra hours and 20 did not. The extra-study group averaged 5 points higher. In 1,000 random regroupings of the 40 scores, a difference of 5 points or more (first group minus second) happened 18 times.

Using a 5% threshold, is the difference significant? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions HSS-IC.B.5 • Set 65 • FORWARD 2 • Q1/5

In a randomized experiment, a tutoring group averaged 6 points more than a control group. Random regroupings of the scores gave differences of 6 points or more 30% of the time.

Using a 5% threshold, what is the conclusion?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 65 • FORWARD 2 • Q2/5

What is the purpose of random assignment in an experiment?

A. To balance other variables between the groups, on average

B. To make the sample smaller

C. To guarantee a significant result

D. To remove the need for a control group

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S65-F2-Q3**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions HSS-IC.B.5 • Set 65 • FORWARD 2 • Q3/5

True or false?

A result is statistically significant when a difference that large would rarely happen by chance alone, using a threshold chosen in advance such as 5%.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-F2-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 65 • FORWARD 2 • Q4/5

True or false?

Using a 5% threshold, a difference that happens in 40% of random regroupings is statistically significant.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 65 • FORWARD 2 • Q5/5

Interpret the slope and the intercept of a linear model in the context of the data

MAIN	Algebra I main question	**HSS-ID.C.7**
BACKWARD 1	Interpret the parameters of linear and exponential functions	**HSF-LE.B.5**
BACKWARD 2	Use a linear model to solve problems	**8.SP.A.3**
FORWARD 1	Interpret the parameters of continuous exponential models	**HSF-LE.A.4**
FORWARD 2	Understand a fitted slope as an estimate of a population value	**HSS-IC.A.1**

SET 66 • HSS-ID.C.7

MAIN

Algebra I main question • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7

$\hat{y} = 4x + 58$ predicts a test score from x hours of study.

Interpret the slope and the intercept.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-M-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7 • Set 66 • MAIN • Q1/5

A model for used cars is $\hat{y} = 22{,}000 - 1{,}800x$, where x is the age in years and $\hat{y}$ is the price in dollars.

Interpret $-1{,}800$ and $22{,}000$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-M-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7 • Set 66 • MAIN • Q2/5

$\hat{y} = 0.9x - 20$ predicts an adult's weight in pounds from height in inches.

Does the intercept make sense in context? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-M-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7 • Set 66 • MAIN • Q3/5

$\hat{y} = 2.5x + 10$ predicts the cost in dollars of a pizza with x toppings.

What does 2.5 mean?

A. Each additional topping adds about \$2.50 to the predicted cost.

B. A pizza has 2.5 toppings.

C. A pizza costs \$2.50.

D. A plain pizza costs \$2.50.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-M-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7 • Set 66 • MAIN • Q4/5

True or false?

The slope of a fitted line gives the exact change in y for every individual.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-M-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7 • Set 66 • MAIN • Q5/5

SET 66 • HSS-ID.C.7

BACKWARD 1

Interpret the parameters of linear and exponential functions • High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5

A technician charges $C = 50 + 20h$ dollars for h hours.

What does 20 mean?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B1-Q1

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 66 • BACKWARD 1 • Q1/5

$P = 3{,}000(0.9)^t$ is a population after t years.

What does 0.9 mean?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B1-Q2

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 66 • BACKWARD 1 • Q2/5

$d = 300 - 60t$ is a driver's distance from home, in miles, after t hours.

What does 300 mean?

A. The starting distance from home

B. The speed

C. The distance driven each hour

D. The time to arrive home

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B1-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 66 • BACKWARD 1 • Q3/5

True or false?

In y = 1.5x + 4, increasing x by 2 increases y by 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B1-Q4

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 66 • BACKWARD 1 • Q4/5

True or false?

In $y = 8(1.25)^x$, y increases by 1.25 for each unit of x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B1-Q5

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 66 • BACKWARD 1 • Q5/5

SET 66 • HSS-ID.C.7

BACKWARD 2

Use a linear model to solve problems • Grade 8 • Statistics & Probability • 8.SP.A.3

$y = 2x + 5$ models a plant's height in cm after x weeks.

Predict the height at 6 weeks.

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B2-Q1

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 66 • BACKWARD 2 • Q1/5

Using $y = 2x + 5$, after how many weeks is the plant 21 cm tall?

Weeks: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B2-Q2

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 66 • BACKWARD 2 • Q2/5

$y = 2x + 5$ models a plant's height in cm after x weeks.

In this model, what does 5 mean?

A. The height at week 0

B. The weeks needed to grow 1 cm

C. The final height

D. The growth per week

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B2-Q3

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 66 • BACKWARD 2 • Q3/5

True or false?

The slope of a linear model is a rate of change.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B2-Q4

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 66 • BACKWARD 2 • Q4/5

True or false?

In y = 2x + 5, the plant grows 5 cm each week.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B2-Q5

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 66 • BACKWARD 2 • Q5/5

SET 66 • HSS-ID.C.7

FORWARD 1

Interpret the parameters of continuous exponential models • High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

$A(t) = 500e^{0.04t}$.

What does 0.04 mean? When does A double, to the nearest tenth?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-F1-Q1

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 66 • FORWARD 1 • Q1/5

$N(t) = 1{,}000e^{-0.1t}$.

What does -0.1 mean? When is $N = 500$, to the nearest tenth?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-F1-Q2

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 66 • FORWARD 1 • Q2/5

Solve $e^{2x} = 9$ exactly. (ln means base e.)

A. $x = \ln 3$

B. $x = 4.5$

C. $x = e^{4.5}$

D. $x = \ln 9 \div 9$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-F1-Q3

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 66 • FORWARD 1 • Q3/5

True or false?

$\ln(e^{0.04t}) = 0.04t$ (ln means base e.)

A. True

B. False

True or false?

If $e^k = 1$, then $k = 1$.

A. True

B. False

SET 66 • HSS-ID.C.7

FORWARD 2

Understand a fitted slope as an estimate of a population value • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1

A random sample of 50 used cars of one model gives $\hat{y} = 22{,}000 - 1{,}800x$.

What population value does $-1{,}800$ estimate?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.A.1 • Set 66 • FORWARD 2 • Q1/5

Would another random sample of 50 cars give exactly the same slope? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions HSS-IC.A.1 • Set 66 • FORWARD 2 • Q2/5

Which of these is a parameter?

A. The mean price of the 50 sampled cars

B. One car's price

C. The slope from the sample

D. The mean price of all used cars of this model

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-F2-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions

HSS-IC.A.1 • Set 66 • FORWARD 2 • Q3/5

True or false?

Larger random samples tend to give statistics closer to the parameter.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-F2-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.A.1 • Set 66 • FORWARD 2 • Q4/5

True or false?

A statistic is a number that describes a whole population.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions HSS-IC.A.1 • Set 66 • FORWARD 2 • Q5/5

Compute (using technology) and interpret the correlation coefficient of a linear fit

MAIN	Algebra I main question	**HSS-ID.C.8**
BACKWARD 1	Describe the direction of an association	**8.SP.A.1**
BACKWARD 2	Fit and use a line for linear data	**HSS-ID.B.6.c**
FORWARD 1	Use a randomized experiment and simulation to test whether an a…	**HSS-IC.B.5**
FORWARD 2	Evaluate claims based on a correlation	**HSS-IC.B.6**

SET 67 • HSS-ID.C.8

MAIN

Algebra I main question • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8

Use technology to find the correlation coefficient r for the data

(1, 9), (2, 8), (3, 8), (4, 6), (5, 5), (6, 5), (7, 3), (8, 2),

to the nearest hundredth. Describe the association.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-M-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 67 • MAIN • Q1/5

$r = -0.85$ for hours of TV and test score.

Interpret this value.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-M-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 67 • MAIN • Q2/5

Which value of r shows the strongest linear relationship?

A. 0.10

B. −0.92

C. −0.40

D. 0.75

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-M-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 67 • MAIN • Q3/5

Can a correlation coefficient equal 1.3? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-M-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 67 • MAIN • Q4/5

True or false?

r = 0 means there is no relationship of any kind between the variables.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-M-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 67 • MAIN • Q5/5

SET 67 • HSS-ID.C.8

BACKWARD 1

Describe the direction of an association • Grade 8 • Statistics & Probability • 8.SP.A.1

Points rise from left to right, fairly close to a line.

Describe the association.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 67 • BACKWARD 1 • Q1/5

Points for hours of sleep and number of errors fall from left to right.

Is the association positive or negative?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 67 • BACKWARD 1 • Q2/5

Which pair of variables likely has a positive association?

A. Height and arm span

B. Outdoor temperature and heating cost

C. A car's age and its value

D. Speed and travel time for a fixed trip

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 67 • BACKWARD 1 • Q3/5

True or false?

In a negative association, as one variable increases, the other tends to decrease.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 67 • BACKWARD 1 • Q4/5

True or false?

Clusters in a scatter plot mean the data are linear.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 67 • BACKWARD 1 • Q5/5

SET 67 • HSS-ID.C.8

BACKWARD 2

Fit and use a line for linear data • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c

Write a line of fit through (2, 5) and (8, 14).

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B2-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c • Set 67 • BACKWARD 2 • Q1/5

The least-squares line is $\hat{y} = 1.5x + 2$.

Predict y when x = 6.

$\hat{y} =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B2-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c • Set 67 • BACKWARD 2 • Q2/5

What does the least-squares line minimize?

A. The y-intercept

B. The sum of the squared residuals

C. The number of points

D. The slope

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B2-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c • Set 67 • BACKWARD 2 • Q3/5

True or false?

The least-squares line passes through the point $(\bar{x}, \bar{y})$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B2-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c • Set 67 • BACKWARD 2 • Q4/5

True or false?

Two people drawing lines of fit by eye always get the same line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B2-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c • Set 67 • BACKWARD 2 • Q5/5

SET 67 • HSS-ID.C.8

FORWARD 1

Use a randomized experiment and simulation to test whether an association is causal • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5

Observational data show r = −0.6 between evening screen time and hours of sleep. To test cause, 50 volunteers were randomly assigned to no screens after 8 p.m. or to their usual habits. The no-screen group slept 0.5 hour more on average. In 1,000 random regroupings, a difference of at least 0.5 hour (no-screen minus usual) happened 2 times.

Using a 5% threshold, is the difference significant? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-F1-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions HSS-IC.B.5 • Set 67 • FORWARD 1 • Q1/5

Why can this randomized experiment, but not the observational correlation alone, support the claim that evening screens reduce sleep?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-F1-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions HSS-IC.B.5 • Set 67 • FORWARD 1 • Q2/5

In a randomization test, the observed difference was matched or exceeded in 350 of 1,000 regroupings.

Using a 5% threshold, what is the conclusion?

A. The treatment definitely has no effect.

B. The difference is not significant; chance could easily explain it.

C. The difference is significant.

D. The experiment proves causation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-F1-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions

HSS-IC.B.5 • Set 67 • FORWARD 1 • Q3/5

True or false?

In a randomization test, the regroupings show how large the difference between groups tends to be when the treatment has no effect.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-F1-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 67 • FORWARD 1 • Q4/5

True or false?

If an experiment's difference is not significant, that proves the treatment has no effect.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-F1-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 67 • FORWARD 1 • Q5/5

SET 67 • HSS-ID.C.8

FORWARD 2

Evaluate claims based on a correlation • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6

A news story says, "r = 0.6, so 60% of the change in sales is caused by advertising."

Name two errors.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 67 • FORWARD 2 • Q1/5

A study of 8 cities finds r = 0.7 between the number of parks and happiness scores.

Why should you be cautious?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 67 • FORWARD 2 • Q2/5

Which statement about $r = -0.9$ is correct?

A. There is a strong negative linear association.

B. The slope of the line is -0.9.

C. 90% of the points are on the line.

D. An increase in x causes y to fall.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S67-F2-Q3**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 67 • FORWARD 2 • Q3/5

True or false?

A correlation can be strong even when the relationship is not causal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-F2-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 67 • FORWARD 2 • Q4/5

True or false?

If r = 0.8, the slope of the least-squares line is 0.8.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions HSS-IC.B.6 • Set 67 • FORWARD 2 • Q5/5

SET 68 • HSS-ID.C.9 • INTERPRETING CATEGORICAL & QUANTITATIVE DATA

Distinguish between correlation and causation

MAIN	Algebra I main question	**HSS-ID.C.9**
BACKWARD 1	Interpret the correlation coefficient	**HSS-ID.C.8**
BACKWARD 2	Use random samples to make valid generalizations	**7.SP.A.1**
FORWARD 1	Distinguish surveys, experiments, and observational studies	**HSS-IC.B.3**
FORWARD 2	Use random methods to make fair decisions — nearest related	**HSS-MD.B.6**

SET 68 • HSS-ID.C.9

MAIN

Algebra I main question • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9

Data show that cities with more firefighters have more fire damage.

Do these data establish that hiring firefighters causes damage? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-M-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 68 • MAIN • Q1/5

Ice cream sales and sunburn rates are strongly correlated.

Name a lurking variable.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-M-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 68 • MAIN • Q2/5

How could you test whether a new study app causes higher test scores?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-M-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 68 • MAIN • Q3/5

Which study design can show that one variable causes changes in another?

A. A survey of volunteers

B. A scatter plot of existing data

C. An observational study

D. A randomized experiment

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-M-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 68 • MAIN • Q4/5

True or false?

A strong correlation between two variables proves that one causes the other.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-M-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 68 • MAIN • Q5/5

SET 68 • HSS-ID.C.9

BACKWARD 1

Interpret the correlation coefficient • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8

$r = 0.88$.

Describe the association.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B1-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 68 • BACKWARD 1 • Q1/5

Which shows the stronger linear association, $r = -0.7$ or $r = 0.5$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B1-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 68 • BACKWARD 1 • Q2/5

What is r for points that lie exactly on a line with negative slope?

A. 1

B. 0

C. −0.5

D. −1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B1-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 68 • BACKWARD 1 • Q3/5

True or false?

The correlation coefficient has no units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B1-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 68 • BACKWARD 1 • Q4/5

True or false?

Switching the roles of x and y changes the value of r.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B1-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 68 • BACKWARD 1 • Q5/5

SET 68 • HSS-ID.C.9

BACKWARD 2

Use random samples to make valid generalizations • Grade 7 • Statistics & Probability • 7.SP.A.1

Why is a random sample more likely to represent a population than a convenience sample?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B2-Q1

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 68 • BACKWARD 2 • Q1/5

A survey at a gym asks about the exercise habits of all town residents.

What is the problem?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B2-Q2

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 68 • BACKWARD 2 • Q2/5

Which is a random sample of a school's students?

A. Drawing 40 names from a list of all students

B. Asking the first 40 students in line

C. Asking your friends

D. Asking one math class

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B2-Q3

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 68 • BACKWARD 2 • Q3/5

True or false?

Generalizations about a population are valid only if the sample represents the population.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B2-Q4

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 68 • BACKWARD 2 • Q4/5

True or false?

A sample of volunteers is a random sample.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B2-Q5

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 68 • BACKWARD 2 • Q5/5

SET 68 • HSS-ID.C.9

FORWARD 1

Distinguish surveys, experiments, and observational studies • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3

Researchers record students' screen time and grades without changing anything.

What kind of study is this?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-F1-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions HSS-IC.B.3 • Set 68 • FORWARD 1 • Q1/5

Researchers randomly assign plants to receive fertilizer or no fertilizer.

What kind of study is this?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-F1-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions HSS-IC.B.3 • Set 68 • FORWARD 1 • Q2/5

What is the main purpose of a survey?

A. To test whether a treatment causes an effect

B. To assign treatments to subjects

C. To control lurking variables

D. To estimate a characteristic of a population from a sample

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REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S68-F1-Q3**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions HSS-IC.B.3 • Set 68 • FORWARD 1 • Q3/5

True or false?

Random selection lets you generalize to a population; random assignment lets you conclude cause and effect.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-F1-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions

HSS-IC.B.3 • Set 68 • FORWARD 1 • Q4/5

True or false?

In an observational study, the researchers decide which treatment each subject gets.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-F1-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions HSS-IC.B.3 • Set 68 • FORWARD 1 • Q5/5

SET 68 • HSS-ID.C.9

FORWARD 2

Use random methods to make fair decisions • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.6

⚑ No standard in a later course directly continues this skill at this step. This branch uses the nearest related Probability & Statistics standard. Connection: Random assignment, which is needed to show causation, is a fair random method.

Describe a fair way to split 20 volunteers into two groups of 10 using random numbers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 68 • FORWARD 2 • Q1/5 • NEAREST RELATED

A coach picks a captain by drawing one name from a hat with 12 names.

What is the probability that a given player is chosen?

Probability: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 68 • FORWARD 2 • Q2/5 • NEAREST RELATED

Which method is fair for deciding who goes first among 4 players?

A. The tallest goes first

B. Each rolls a number cube; ties roll again; the highest goes first

C. The coach's favorite goes first

D. Whoever asks first goes first

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S68-F2-Q3**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 68 • FORWARD 2 • Q3/5 • NEAREST RELATED

True or false?

A fair lottery gives every participant the same probability of being chosen.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-F2-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 68 • FORWARD 2 • Q4/5 • NEAREST RELATED

True or false?

Flipping one fair coin once is a fair way to choose one person from 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 68 • FORWARD 2 • Q5/5 • NEAREST RELATED